

Every unit justified before the final order.

LastBuy verifies one discontinued image-processing ASIC purchase for an industrial thermal-camera manufacturer.

Decision: how many components must the OEM buy before the supplier closes final orders, while preserving the approved repair obligations for eligible camera cohorts?

Accountability	Exact enterprise role
Economic buyer	Director of Global Service Supply Chain; finance controller co-signs material commitments.
Daily user	Service-parts planner / obsolescence manager reconciling the final-buy proposal.
Approval owners	Sustaining engineer, service-contract owner, finance controller and procurement manager; four distinct people.

Financial stakes with a traceable example

Reported stock can include duplicate custody records, incompatible board revisions and quarantined units. Service amendments can also increase demand. A wrong input becomes unnecessary capital commitment or an uncovered service obligation.

Synthetic decision bridge	Units / value
Reported on-hand stock	14,000
Remove duplicate / incompatible / quarantine	2,000 / 3,000 / 1,000
Eligible stock + confirmed inbound	8,000 + 4,000
Protected demand / required new buy	22,000 / 10,000
Original proposal / verified plan	18,000 x \$80 = \$1.44m / 10,000 x \$80 = \$800,000

\$640,000 modeled commitment difference. The figures are constructed test data, not achieved savings. Real Foundry hosted agents completed this synthetic cloud case. The recommendation can increase when coverage or reservations change.

Business basis: manufacturer discontinuation notices, component-specific warranty terms and material inventory exposures support the problem category. They do not prove that a named customer has validated LastBuy. Sources and limitations: page 6.

Specialists with a reason to exist.

Use language models to reconcile source meaning. Keep quantity, authority and external writes under deterministic control.

Specialist	Evidence and bounded responsibility	Output checked by server
Engineering	Released PLM records; component/board revision and cohort applicability.	Release/effectivity checks, exact quotes and unresolved conflicts.
Service	Signed coverage records and approved demand scenarios; distinguish detector warranty from ASIC coverage.	Coverage and demand-source checks; governing evidence and blockers.
Supply	ERP/depot lot records, reservations, holds and inbound confirmations.	Quantity/confirmation checks; evidence for each disjoint exclusion.
Commitment	Supplier quote, purchase limits and the deterministic calculation.	Price/allocation checks and explanation of the verified commitment.

Implemented tool contract

Each role uses **read_case_evidence** with a server-selected source scope. The tool returns the immutable snapshot hash, permitted source records, required extractions and relevant structured facts. No arbitrary SQL, URL fetch, shell, approval or ERP-write tool is available.

Structured handoffs

Durable activities share a case/run ID and the same accepted snapshot. Each assessment returns role, summary, findings, exact supporting quotes and critical-field checks. The server validates scope, quote membership and agreement with accepted fields; unresolved review findings block approval. The final plan includes allocations, costs, blockers, source hash, policy version and plan hash.

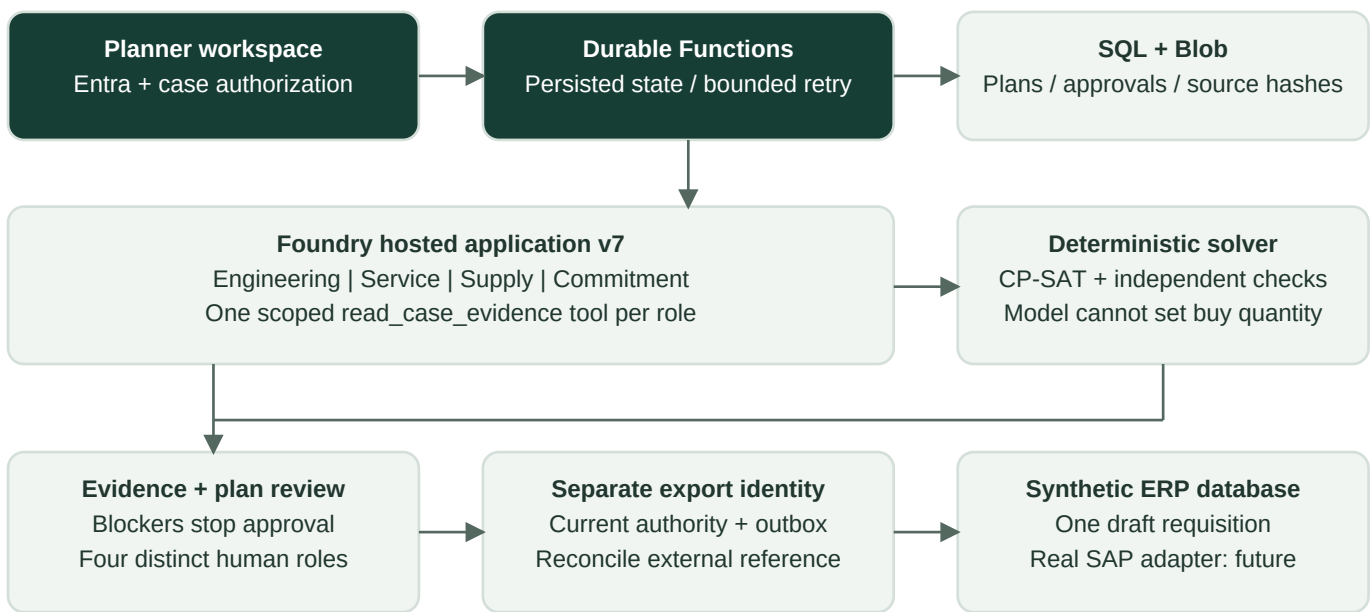
Why multiple agents can help

The engineering, contract and stock records have different owners and authority rules. Scoped assessments make failures attributable and allow role-specific tests and bounded retries. One generic instruction set would combine these authority questions. The fourth role explains supplier commitment against the deterministic result; it does not calculate the purchase.

First matched-input comparison: on the same golden snapshot and model deployment, the four-stage record correctly accepted the case (12,569 response-reported tokens). A single-call comparator using the same four validation contracts falsely flagged resolved duplicate custody (8,363 tokens). This is one case with a historical comparator, not a broad superiority claim. Logical roles share one hosted process.

A decision workflow with durable authority.

Deployed development architecture. Azure services are real; enterprise sources and requisitions are synthetic.



Application Insights: correlated stage/model/tool traces; source content capture disabled.

Sequence from evidence to draft requisition

1. Import a typed, hash-checked synthetic snapshot and source revision.
2. Validate source-owner references; run four bounded specialist activities.
3. Validate assessments; solve and independently verify allocation constraints.
4. Archive the snapshot and store the versioned decision for evidence review.
5. Collect engineering, service, finance and procurement approval for the exact hash.
6. Recheck stored source/policy/authority; enqueue one export reference.
7. The isolated worker writes a synthetic draft and reconciles uncertain responses.

Boundary	Implemented behavior / production extension
Model hosting	Foundry hosted v7; Microsoft Agent Framework; lastbuy-dev-mini model deployment. Deterministic stage order controls current quota and cost.
Data ownership	SAP/depot: stock; Windchill: applicability; Contracts/Dataverse: coverage; Planning: scenarios; Supplier: quote. Current imports trust manifest labels.
Customer integration	Approved SAP/PLM/Dataverse APIs or extracts, complete paging and live revision refetch are required for a customer deployment; not implemented as live connectors.

Make the consequence visible to the planner.

The central interaction is a purchase decision: inspect what changed, which evidence governs it, and who must act.

Workspace surface	User question answered
Quantity bridge	Why does reported stock differ from usable supply? Open the exact source behind an exclusion.
Scenario coverage	Which approved demand envelope is protected? Show allocation per scenario without summing alternatives.
Blockers and evidence	What is unresolved, who owns it, and which source revision must be corrected?
Approval and audit	Which person approved which plan? Are source and authority still current?
Requisition receipt	Was the draft accepted? Can a retry recover the existing reference without another write?

The decisive demonstration

Show the verified 10,000-unit plan. Then introduce a 1,000-unit stock reservation on a separate prepared case. The old approval must become stale and export must stop. The deterministic revised requirement is 11,000 units, subject to fresh analysis and approval. Finish by reconciling a lost-response synthetic export to its original receipt.

Three-minute presentation	Proof to show
0:00-0:25	One buyer, the final-order deadline and the \$1.44m proposal.
0:25-1:15	Coverage clause and stock bridge; the 22,000-unit obligation remains protected.
1:15-2:05	The \$800,000 plan; changed evidence invalidates authority and blocks export.
2:05-2:35	Four visibly synthetic approval roles and one reconciled draft reference.
2:35-3:00	Actual trace, one failure-driven refinement and the measured pilot target.

Use clearly labeled completed Foundry results if a cold run exceeds the recording window. Do not present deterministic previews, test doubles or edited timing as new live agent execution. The participant records and edits the final video.

Prove quality. Expose the failures.

Evaluation and operating evidence are part of the product design, not a final presentation decoration.

Evidence	What it establishes / limit
147 automated tests	Solver, workflow, authority, replay, cancellation, expiry and export behaviors within covered fixtures. Includes regressions from captured provider failures.
60/60 offline suite	50 structured solver fixtures and 10 security regression references. Not a full independent language-understanding benchmark.
New live stock challenge	The model chose 6,000 opening units despite an explicit 4,800 current balance. Captured-response regression drove a currentness gate. The corrected deployed worker rejected a legacy-approved fixture with HTTP 409 and zero ERP rows.
v7 regression: 9 attempts	Six correct accepted outcomes, three abstentions. Four conflicts: one explicit block, three abstentions, zero silent passes. Four clean cases accepted. Reused post-tuning cases; not a new holdout.
Real recovery evidence	Worker kill/resume; actual v2 checkpoint-to-approval completion; real SQL concurrency; cloud export/reconciliation; local backup/restore. Connection timeout evidence drove bounded connection retry.

Evaluation in the development lifecycle

Freeze cases and labels before prompt changes. Run offline arithmetic/authority tests on every relevant change. Within an approved allowance, compare candidate and current hosted versions on an untouched conflict/clean/injection set; retain failed attempts. Gate release on zero observed unsafe exports, exact quantity checks, source resolvability and explicit conflict/abstention metrics. Report sample sizes and false blocks.

Observability that answers operational questions

Correlate case/run, snapshot and plan hashes across Durable, model and tool spans. Track stage duration, retries, token usage, validation failures, approval transitions and uncertain exports. Input/output content capture is disabled. A successful cloud case reported 12,569 stage tokens and 18.085 seconds of model/tool runtime; this excludes end-to-end overhead and is not p95.

A credible path from prototype to production.

The contest entry demonstrates architectural judgment. A real customer rollout needs additional evidence and authority.

Control or outcome	Current implementation and release gate
Identity and approval	Entra PKCE/JWT; current server-owned roles/amount limits; four distinct approvers. Real cloud user is planner-only; demo personas do not establish customer approval.
Safe boundaries	LLM has read-only evidence tools. Separate exporter identity/database; external-reference uniqueness. Supplier text cannot grant engineering authority.
Reliable changes	Source/policy changes invalidate approvals. Cancellation blocks late completion; v2 approval wait expiry is authoritative. V1 is retained for existing replay histories.
Production hardening	Add trusted customer connectors and live source refetch, workload-token export transport, reviewed private networking, locked retention and tested cloud restore/alerts.
Pilot outcome	At least 10 buyer-provided historical cases. Target at least 50% less active reconciliation time, no increased missed coverage and positive finance-validated net benefit. Targets are not measured results.

Differentiation to test

Servigistics already provides last-time-buy planning; Z2Data already offers obsolescence cases, integration and approvals. LastBuy targets verification across component-specific service obligations, released engineering effectivity and uniquely owned eligible stock, binding the resulting commitment to fresh approval and a recoverable export. Public product descriptions cannot establish exclusive capability or competitive superiority.

Selected sources and reproducible evidence

Microsoft Agent-a-Thon / Level 3

Official rules / judging and entry requirements

Authenticated Architect final assignment

Microsoft teaching repository

Microsoft orchestration guidance

Teledyne Q2 2026 filing / inventory context

FLIR component-specific warranty

PTC Servigistics capabilities

Z2Data lifecycle management

Source review: 20 September 2026. Repository evidence: full-cloud-release7.json; evaluation-summary.json; implementation-tests.xml; cloud-export-verification.json; final-release-check.json. Full source qualifications and competitor links are in docs/01-business-case.md.